

Definition. The function $z = f(x_1, x_2, \dots, x_n)$ is **linear** if there are multipliers $p_1, p_2, \dots, p_n$ for which

$$\Delta z = p_1 \cdot \Delta x_1, \quad \Delta z = p_2 \cdot \Delta x_2, \quad \dots, \quad \Delta z = p_n \cdot \Delta x_n.$$